

The STARDUST project: investigating altered motion perception and its consequences in PPPD

Andrew P. Jones^{1,2}, Joshua D. Haynes¹, Boris Otkhmezuri¹, Graham Bell¹, Debbie Cane^{1,3}, Rosa Crunkhorn⁴, Monty Silverdale², James B. Lilleker², Alan Carson⁵, Paul A. Warren¹

¹University of Manchester, UK, ²Manchester Centre for Clinical Neurosciences, UK, ³Audiology, Manchester University NHS Foundation Trust, UK, ⁴Audiovestibular Medicine, Guy's and St Thomas' NHS Foundation Trust, London, UK, ⁵Centre for Clinical Brain Sciences, Edinburgh, UK

1. Why study motion perception in PPPD?

- Persistent Postural-Perceptual Dizziness (PPPD) is a chronic functional vestibular disorder characterised by persistent dizziness, unsteadiness and visually-induced symptoms¹
- The STARDUST project investigates whether PPPD involves altered processing of self-motion signals within sensorimotor networks

Two mechanistic hypotheses

Mismatch

Symptoms arise from inappropriate compensation for reafferent visual signals

Integration

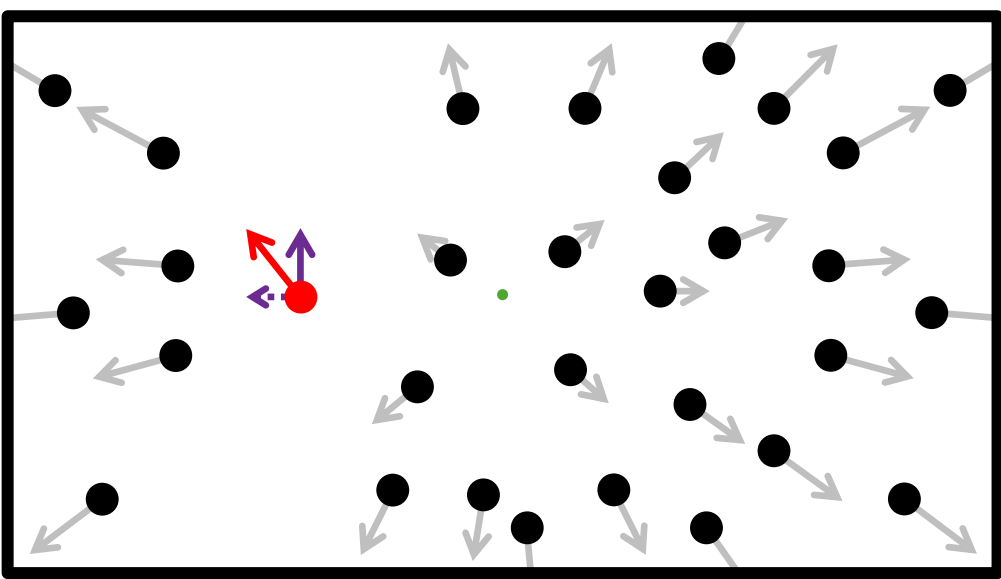
Symptoms arise from suboptimal combination of visual and vestibular input (compared to near-optimal in healthy controls²)

Posturography hypotheses

- Force-plate posturography studies of functional dizziness disorders have identified transition to a stiffened postural control strategy³
- We sought to assess postural control in PPPD via motion capture and pre-registered hypotheses that greater sway (vs. healthy controls) would diminish with more difficult stance and visual input condition (*AsPredicted* # 250,803)
- Unlike past work on postural misperception, we predicted that observed postural sway would differ between groups rather than being purely perceived as different⁴

2. STARDUST Experiment pathway

Optic flow parsing



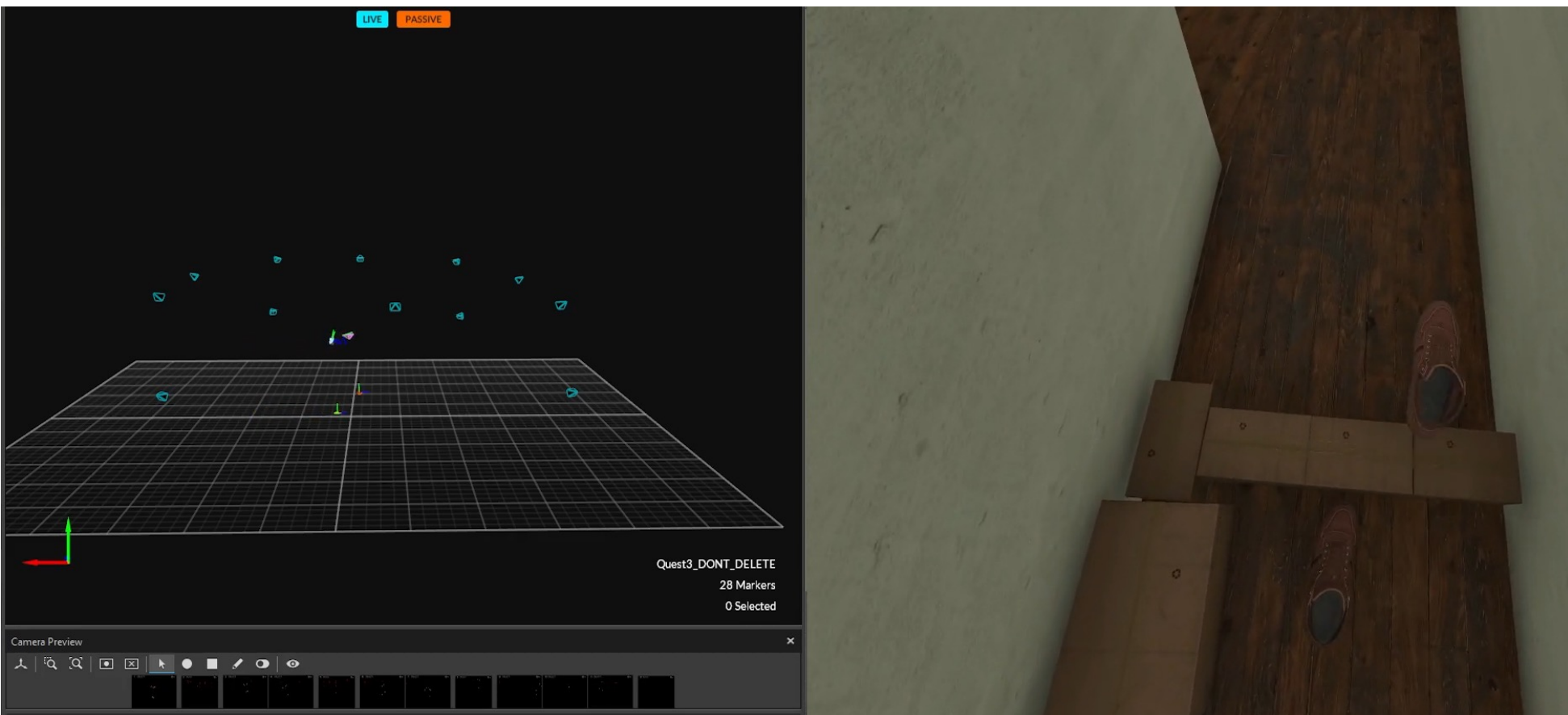
Question: Is compensation for visual self-motion altered?

Measure: flow parsing bias and discrimination thresholds

VR Walking task

Question: How does altered perception affect walking?

Measure: gait stability and obstacle collisions



Visual-vestibular integration



Question: Are visual and vestibular cues combined optimally?

Measure: heading and scene relative object movement thresholds and integration score

3. Quiet posturography - Methods

- Symptom characteristics assessed via Dizziness Handicap Inventory (DHI) and General Anxiety Disorder-7 (GAD-7) questionnaires
- Quiet stance posturography in different stance (Romberg, tandem) and visual input (eyes open/closed) conditions (as shown in Fig. 1)
- Head kinematic time-series (20 s) recorded via marker-based motion capture, used to derive sway metrics, including sway area (95% prediction ellipse) from head trajectories (as shown in Fig. 2)
- Here we present an interim analysis
- N = 33 (21 patients, PPPD, 12 healthy controls, HC)

Variable	HC (n=12)	PPPD (n=21)	P value
Age, mean (SD)	48.42 (14.93)	50.14 (11.63)	.734
Sex, n (%) Female	5 (41.7%)	15 (71.4%)	.092
DHI, median (IQR)	0 (0.0)	62 (18.0)	< .001
GAD-7, median (IQR)	2.0 (3.5)	3.0 (6.0)	.105

- Statistical comparison of log-transformed ellipse areas in a 2x(2x2) mixed ANOVA design
- Spearman rank coefficient correlation of 95% prediction ellipse areas with DHI scores

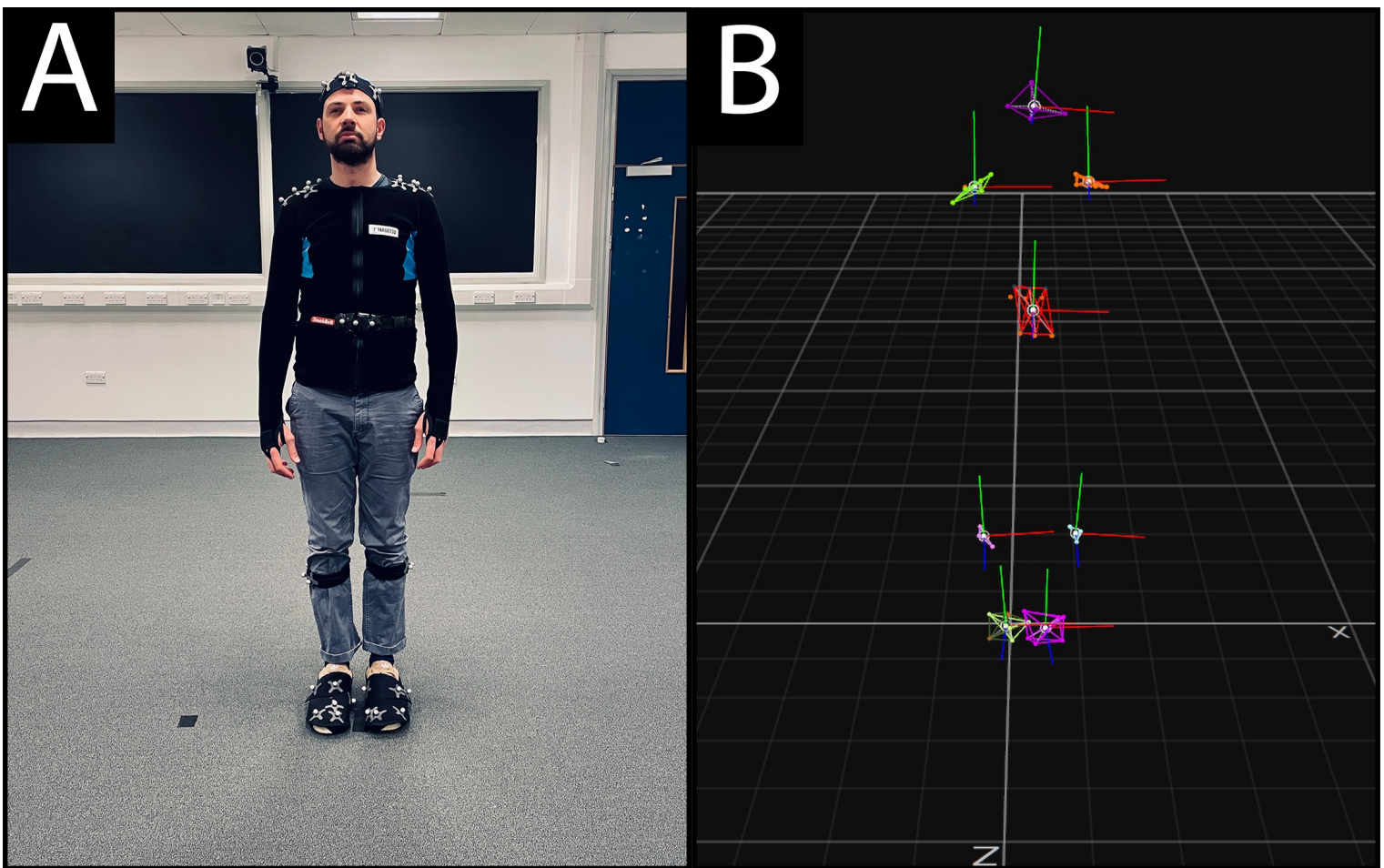


Figure 1. A: Marker-based motion capture setup. B: Markers reconstructed in Motive software for postural sway recording.

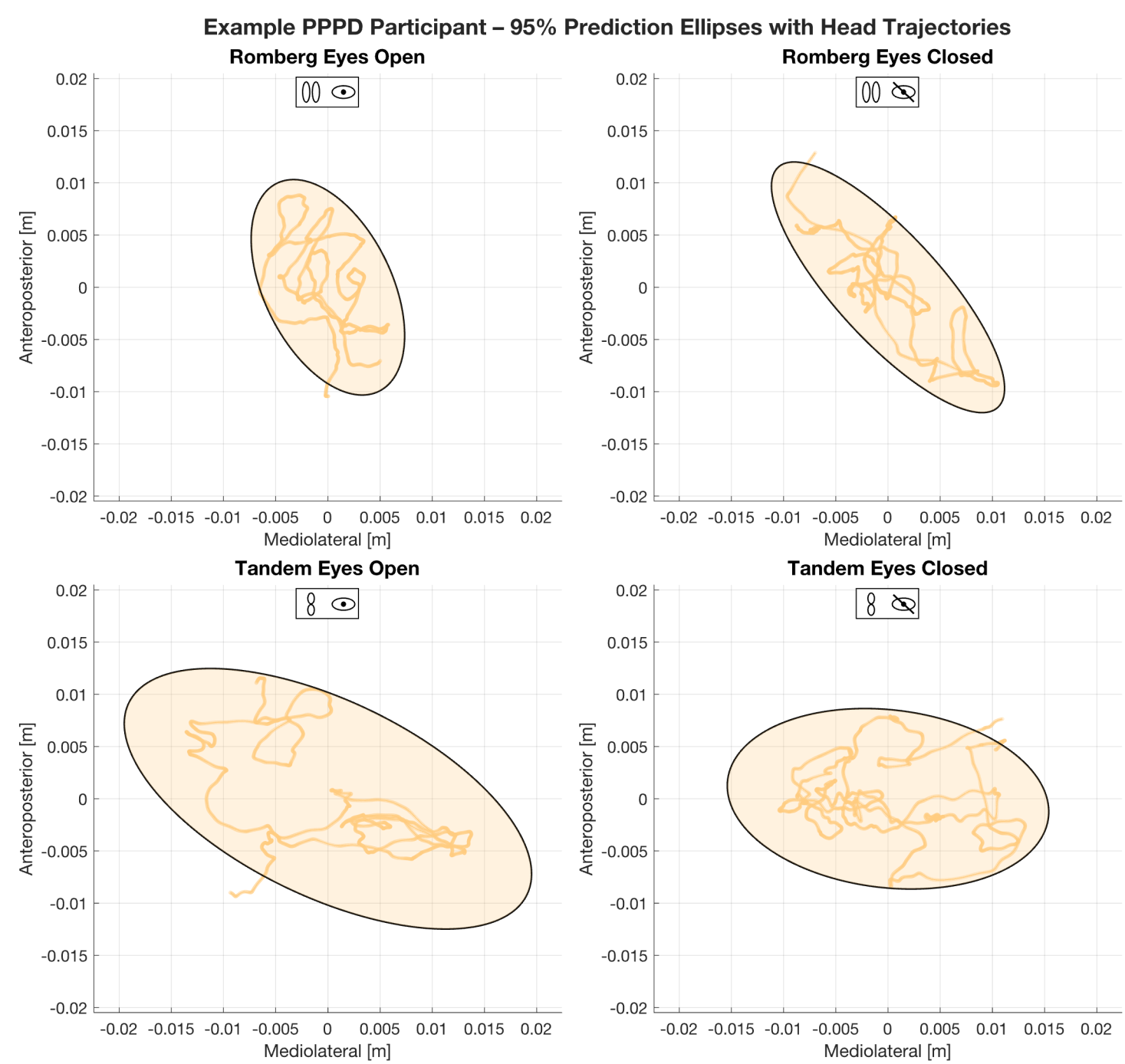


Figure 2. Head trajectories during quiet stance posturography with overlaid 95% prediction ellipses.

4. Posturography - Results

- Patients exhibited greater postural sway than controls, with sway increasing under more demanding balance conditions
- However, in the most challenging condition, PPPD group showed lower sway than controls, potentially reflecting maladaptive postural stiffening (Fig 3.)

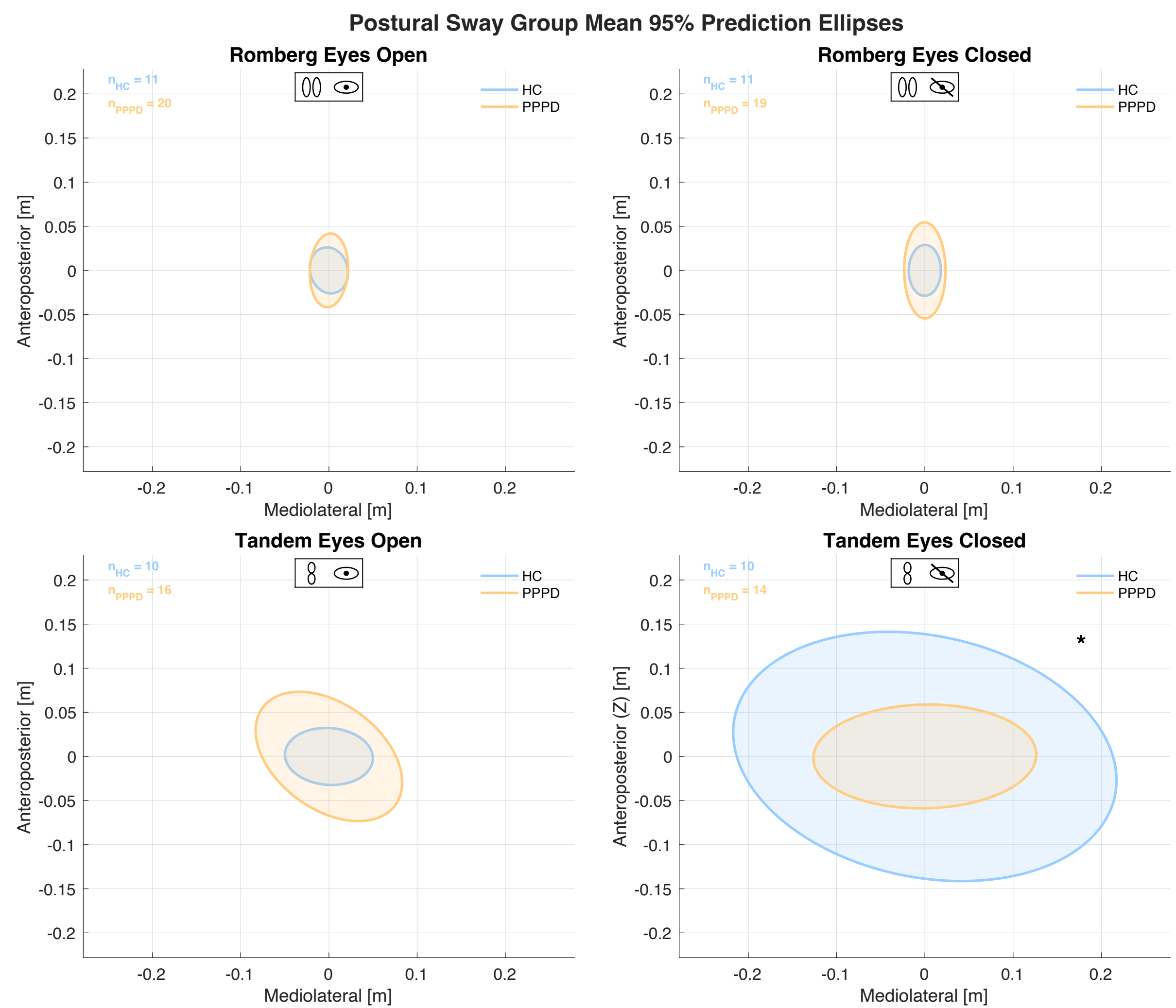


Figure 3. Mean 95% prediction ellipse area by group and condition.

*Condition contributing to the significant Group x Stance x Visual Input interaction, ($F(1,20) = 6.67$, $p = .018$, $\eta^2 = .25$), indicating that sway responses to sensory conditions differed between groups.

In the tandem eyes-closed condition, mean sway area was $0.066 \pm 0.036 \text{ m}^2$ in HC and $0.020 \pm 0.013 \text{ m}^2$ in PPPD.

- Dizziness-related disability (as measured by DHI) showed strong positive correlation with sway area in the PPPD group:

Romberg Eyes Open	$\rho = .71$, $p = .001$ ($n = 18$)
Romberg Eyes Closed	$\rho = .68$, $p = .002$ ($n = 17$)
Tandem Eyes Open	$\rho = .47$, $p = .076$ ($n = 15$)
Tandem Eyes Closed	$\rho = .57$, $p = .040$ ($n = 13$)

5. Discussion

- Interim results find evidence for altered postural control in PPPD consistent with maladaptive stiffening, suggesting sensorimotor network dysfunction
- Lower PPPD group sway in most challenging condition may reflect calf muscle co-contraction as found previously in EMG study of phobic postural vertigo⁵
- Perceived dizziness-related disability was positively associated with objectively measured sway, which may be inconsistent with previous work emphasising postural misperception in PPPD⁴
- Future work will explore therapeutic approaches using manipulations of the VR visual environment

REFERENCES

1. Staab et al. (2017). J Vestib Res, 27. 2. Yakubovich et al. (2020). Brain Commun, 2. 3. Krafczyk et al. (1999). Neurosci Lett, 259. 4. San Pedro Munillo et al. (2023). JNNP, 94. 5. Wuehr et al. (2017). Neurology, 88.neurot